AVD Day-to-Day Management

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Managing an active Azure Virtual Desktop (AVD) environment requires a different set of commands than the initial deployment. Below are practical PowerShell workflows for day-to-day AVD administration.

Ensure you have authenticated to Azure (`Connect-AzAccount`) and selected the correct subscription before running these commands. We will assume your variables (`$rgName`, `$hostPoolName`, etc.) are already defined from the previous deployment steps.

Enable Drain Mode for Maintenance

Before patching or restarting a session host, you must prevent new users from connecting to it. Enabling drain mode stops new connections while allowing existing users to finish their work.

```
$sessionHostName = "YOUR-SESSION-HOST-NAME"

Update-AzWvdSessionHost -ResourceGroupName $rgName `
  -HostPoolName $hostPoolName `
  -Name $sessionHostName `
  -AllowNewSession:$false
```

Publish a Specific Application (RemoteApp)

If you created a RemoteApp Application Group instead of a Desktop Group, you must define which applications users can launch. You need the exact file path of the executable on the session host.

```
$remoteAppGroupName = "Finance-RemoteApp-DAG"
$appName = "Excel"
$appPath = "C:\Program Files\Microsoft Office\root\Office16\EXCEL.EXE"

New-AzWvdApplication -ResourceGroupName $rgName `
  -GroupName $remoteAppGroupName `
  -Name $appName `
  -FilePath $appPath `
  -CommandLineSetting DoAllow
```

Identify Active User Sessions

To troubleshoot a user issue or prepare for maintenance, you often need to find out exactly where a user is logged in.

```
$userUPN = "jdoe@contoso.com"

Get-AzWvdUserSession -ResourceGroupName $rgName -HostPoolName $hostPoolName |
  Where-Object { $_.UserPrincipalName -eq $userUPN } |
  Select-Object UserPrincipalName, SessionHostName, SessionId, SessionState
```

Send a Message to a User

If you need to reboot a server, it is best practice to warn active users first so they can save their work. You need the specific Session ID and Session Host Name from the previous command.

```
$targetSessionHost = "YOUR-SESSION-HOST-NAME"
$targetSessionId = "1"
$title = "IT Maintenance Warning"
$message = "Please save your work and log off. This server will reboot in 15 minutes."

Send-AzWvdUserSessionMessage -ResourceGroupName $rgName `
  -HostPoolName $hostPoolName `
  -SessionHostName $targetSessionHost `
  -SessionId $targetSessionId `
  -MessageTitle $title `
  -MessageBody $message
```

Forcefully Log Off a User

If a user session is hung, or the maintenance window has arrived and users ignored the warning message, you can forcefully log them out.

```
$targetSessionHost = "YOUR-SESSION-HOST-NAME"
$targetSessionId = "1"

Remove-AzWvdUserSession -ResourceGroupName $rgName `
  -HostPoolName $hostPoolName `
  -SessionHostName $targetSessionHost `
  -SessionId $targetSessionId
```

Remove a Session Host from a Host Pool

When scaling down or replacing a broken VM, you must officially unregister it from the host pool before deleting the VM resource in Azure.

```
$sessionHostName = "YOUR-SESSION-HOST-NAME"
```

```
Remove-AzWvdSessionHost -ResourceGroupName $rgName `
  -HostPoolName $hostPoolName `
  -Name $sessionHostName `
  -Force
```

While PowerShell is excellent for automation, the Azure Portal provides a highly visual, structured wizard that is often preferred for initial deployments and ongoing visual management. Here is a practical, step-by-step guide to deploying and managing Azure Virtual Desktop (AVD) using the Azure Portal, along with common portal-based use cases.

Deploying AVD via the Azure Portal

The portal streamlines the creation of the Host Pool, Workspace, and Application Group into a single, cohesive workflow.

Step 1: Create the Host Pool

1. Sign in to the **Azure Portal** (portal.azure.com).
2. In the global search bar, type **Azure Virtual Desktop** and select the service.
3. In the left-hand menu, select **Host pools**, then click **+ Create**.
4. **Basics Tab:**
   - **Resource group:** Create a new one (e.g., `AVD-RG`).
   - **Host pool name:** Name your pool (e.g., `Finance-Pooled-HP`).
   - **Location:** Select the region for your metadata (e.g., `East US`).
   - **Validation environment:** Select **No** (unless this is explicitly a test ring).
   - **Host pool type:** Select **Pooled** (for shared VMs) or **Personal** (for dedicated VMs).
   - **Load balancing algorithm:** Select **Breadth-first** (for performance) or **Depth-first** (for cost optimization).

Step 2: Provision Session Hosts (Virtual Machines)

You can deploy your virtual machines immediately within the same wizard.
1. Click **Next: Virtual Machines**.
2. Select **Yes** under "Add virtual machines".
3. **Image:** Choose your operating system. For a pooled environment, **Windows 11 Enterprise multi-session** is standard.
4. **Virtual machine size:** Select the hardware profile based on your expected user workload (e.g., `Standard_D4s_v5`).
5. **Network and security:** Select your existing Virtual Network and Subnet. Ensure this VNet has line-of-sight to your Active Directory or Microsoft Entra ID.
6. **Domain to join:** Select **Microsoft Entra ID** or **Active Directory**, and provide the required domain administrator credentials.

Step 3: Configure Workspace and App Groups

1. Click **Next: Workspace**.
2. Under "Register desktop app group", select **Yes**.
3. **To this workspace:** Select **Create new**, name the workspace (e.g., Corporate-Workspace), and click **OK**.
4. Click **Review + create**, validate your settings, and click **Create**. Azure will now build the infrastructure and deploy the VMs.

2. Assigning Users in the Portal

Once the deployment completes, the infrastructure exists, but users cannot see it until you explicitly grant them access.

1. Navigate back to the **Azure Virtual Desktop** service page.
2. Select **Application groups** from the left menu.
3. Click on the default application group that was just created (usually named [HostPoolName]-DAG).
4. In the left menu of the application group, select **Assignments**.
5. Click **+ Add**, search for your Entra ID users or security groups, and click **Select**. Your users can now log into the AVD client.

3. Common Portal Use Cases

Administrators frequently rely on the Azure Portal for visibility, troubleshooting, and cost management tasks that are highly visual.

Use Case A: Shadowing and Session Troubleshooting

When a user reports a frozen application or connection issue, the portal provides quick visibility.

- **Action:** Go to your **Host pool**, select **Users**, and search for their email address. The portal will display their active session, the specific VM they are connected to, and their connection state. You can click **Disconnect** or **Log off** directly from this screen to clear hung sessions.

Use Case B: Managing Host Pool Scaling (Cost Optimization)

Leaving VMs running 24/7 is expensive. The portal integrates deeply with Azure Automation to manage power states.

- **Action:** Using the **Scaling Plans** feature within the AVD portal, you can define schedules. For example, you can instruct Azure to ensure 10 VMs are running at 8:00 AM on Monday, but aggressively power down empty VMs to just 2 running hosts after 6:00 PM.

Use Case C: Publishing Individual Apps (RemoteApp)

If a department only needs a specific legacy application, you do not need to give them a full desktop.

- **Action:** Create a new Application Group, select **RemoteApp** instead of Desktop, and link it to your Host Pool. The portal will automatically scan the Start Menu of your session hosts and present you with a clickable list of installed software (like Word, SAP, or Chrome). Select the app, assign your users, and they will only see that specific application in their portal.

If you are looking for a visual walkthrough of the interface and conditional access policies, this walks through the exact setup interface.